Practice workbook · Compare sorts and searches

How linear search, binary search, bubble sort and merge sort work in words, and their relative advantages, drawbacks, memory use and suitability for a given data set.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T1-11

1. Question 3 · part b · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part b only.

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3 A youth club organises age-specific events for its members.

A program is used to find members of the correct age.

(a) The program first sorts the list of members by age.

(i) A bubble sort algorithm is used to sort the list into ascending order.

The table shows a list of ages that must be sorted.

Complete the table to show how the algorithm sorts the list.

You may not need to use all the rows.

(2)

9	11	12	19	14	18	15	17

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(ii) State the number of swaps made in the first pass.

(1)

(iii) State the **least** number of passes needed to complete the sort algorithm for this list.

(1)

1. Question 3 · part b · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part b only.

(b) The youth club stores a list of sports equipment.

The list contains over 1000 items, sorted in ascending order.

State and justify the **most** appropriate search algorithm to find an item in the list.

(2)

Search algorithm

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Justification

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1. Question 3 · part b · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part b only.

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(c) A program is required to calculate and display the average age of members taking part in an activity.

Figure 3 shows a flowchart for the algorithm.

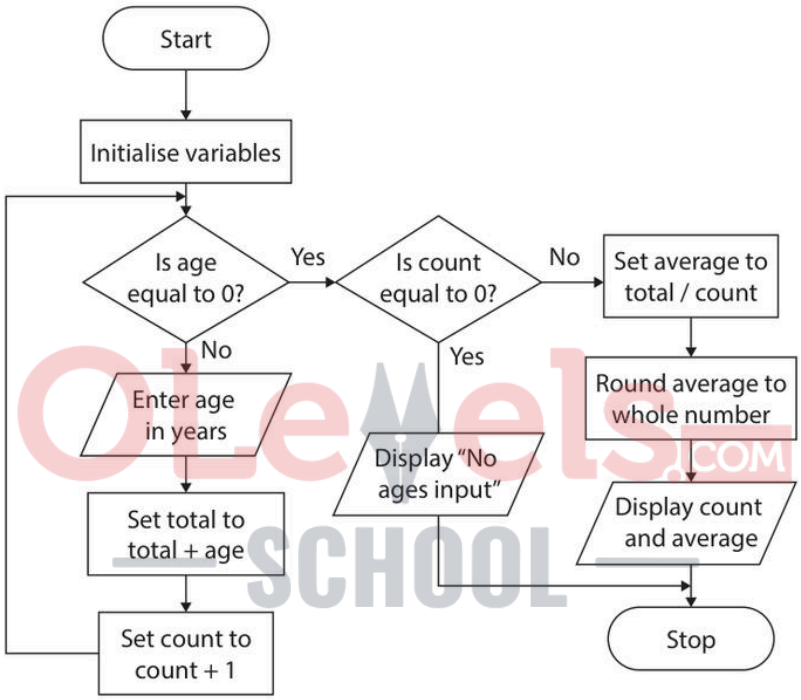


Figure 3

Open **Q03c** in the code editor.

Write a program to implement the logic of the flowchart.

Use the data given in **Figure 4** to test your program.

Test data	Expected output
7 8 9 10 11 0	You entered 5 ages; the average age is 9.
16 16 14 0	You entered 3 ages; the average age is 15.
0	No ages input

Figure 4

Do not add any further functionality.

Save your code as **Q03cFINISHED** with the correct file extension for the programming language.

(5)

(Total for Question 3 = 11 marks)

1. Question 3 · part b · 2 marks · Working and answer

Working and answer

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2. Question 3 · part b · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part b only.

3 Different algorithms can be used to manipulate data.

(a) **Email.txt** contains a list of email addresses.

Open **Q03a** in the code editor.

Write a program to implement these requirements.

The code must:

- check each email address to ensure it contains the '@' symbol.
- write email addresses that do not contain the '@' symbol to an **Error.txt** file.

You must use the structure given in the file **Q03a** to complete the program.

Do not add further functionality.

Save your code as **Q03aFINISHED** with the correct file extension for the programming language.

(6)

(b) Explain **one** drawback of using the merge sort algorithm to sort large data sets.

(2)

(c) Open **Q03c** in the code editor.

Write a program to display the square and cube of a number between 1 and 50 entered by a user.

The code must:

- ask the user to enter a number between 1 and 50 inclusive
- display the number, the square of the number and the cube of the number, with appropriate labels
- stop when a number outside the range 1 to 50 is entered.

Save your code as **Q03cFINISHED** with the correct file extension for the programming language.

(6)

(Total for Question 3 = 14 marks)

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2. Question 3 · part b · 2 marks · Working and answer

Working and answer

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3. Question 5 · part b.i · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part b.i only.

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5 Cerys Jones is the manager of a shop that sells electrical goods.

She is creating a stock control program.

(a) Each item of stock will have a ten-character code.

Each code consists of three parts.

Part	Description
Characters 1 to 3	Uppercase letters
Characters 4 to 9	Numbers between 1 and 9 inclusive
Character 10	'e' or 'o' indicating whether the sum of the numbers is even or odd

WAS123456o, FDG232324e and VAC129222e are valid codes.

All codes need to be validated.

Complete the table to show **two** additional validation tests.

For each test, give **one** example of erroneous data. The example that you give should fail **only** that test.

(4)

Validation test	Erroneous data
Is the last character 'o' or 'e'?	ABC654321v

(b) Cerys will need an algorithm to search the table of stock data.

(i) State **two** advantages of using a linear search algorithm rather than a binary search algorithm.

(2)

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2 .....

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P 6 1 8 8 5 A 0 9 1 6

3. Question 5 · part b.i · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part b.i only.

(ii) State **two** disadvantages of using a linear search algorithm rather than a binary search algorithm.

(2)

1 .....

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2 .....

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(c) Cerys will use functions and procedures in her program.

(i) One similarity between a function and a procedure is that they are both subprograms.

State **two** other similarities.

(2)

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2 .....

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(ii) Explain **one** difference between a function and a procedure.

(2)

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(Total for Question 5 = 12 marks)

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3. Question 5 · part b.i · 2 marks · Working and answer

Working and answer

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4. Question 5 · part b.ii · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part b.ii only.

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Is the last character 'o' or 'e'?	ABC654321v

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(i) State **two** advantages of using a linear search algorithm rather than a binary search algorithm.

(2)

1 .....

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2 .....

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4. Question 5 · part b.ii · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part b.ii only.

(ii) State **two** disadvantages of using a linear search algorithm rather than a binary search algorithm.

(2)

1 .....

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2 .....

.....

(c) Cerys will use functions and procedures in her program.

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State **two** other similarities.

(2)

1 .....

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2 .....

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(ii) Explain **one** difference between a function and a procedure.

(2)

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(Total for Question 5 = 12 marks)

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4. Question 5 · part b.ii · 2 marks · Working and answer

Working and answer

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5. Question 4 · part b.i · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part b.i only.

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4 Zak sells snacks at an after-school club.

(a) He wants a program that will hold details of the products he sells.

The program must:

- generate a five-character product code that:
  - starts with the first three letters of the product name
  - ends with a random number between 10 and 30
- display the product code followed by the product name.

Open **Q04a** in the code editor.

Write the program.

You must use the structure given in the file **Q04a** to complete the program.

Do not add any further functionality.

Save your code as **Q04aFINISHED** with the correct file extension for the programming language.

(7)

(b) Zak plans to implement a binary search algorithm to search a table of products.

(i) Explain **one** advantage of a binary search compared to a linear search.

(2)

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5. Question 4 · part b.i · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part b.i only.

(ii) **Figure 2** shows an algorithm for a binary search.

```
1     #     Initialise variables
2     SET productSearch TO " "
3     SET startPosition TO 1
4     SET midPosition TO 0
5     SET endPosition TO LENGTH(productList)
6     SET found TO False
7
8     #     Print prompts, take and check input from user
9     SEND "Enter the product code" TO DISPLAY
10    RECEIVE productSearch FROM (STRING) KEYBOARD
11    REPEAT
12          midPosition = (startPosition + endPosition) DIV 2
13          IF productList[midPosition] < productSearch THEN
14                startPosition = midPosition + 1
15          END IF
16          IF productList[midPosition] > productSearch THEN
17                endPosition = midPosition – 1
18          END IF
19    UNTIL productList[midPosition] = productSearch OR startPosition = endPosition
```

Figure 2

This binary search algorithm will be used to search the product list for the product code **str15**.

Complete the table to indicate the order in which the product codes will be examined by the algorithm.

Write the number 1 by the first product code to be examined, 2 by the second code to be examined, and so on.

(4)

Position in list	Product code	Order examined
1	ark11	
2	asp11	
3	bar13	
4	dri15	
5	mil19	
6	rib10	
7	str15	
8	tor16	

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P 6 6 4 9 0 A 0 1 2 1 6

5. Question 4 · part b.i · 2 marks · Working and answer

Working and answer

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6. Question 4 · part b.iv · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part b.iv only.

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  - starts with the first three letters of the product name
  - ends with a random number between 10 and 30
- display the product code followed by the product name.

Open **Q04a** in the code editor.

Write the program.

You must use the structure given in the file **Q04a** to complete the program.

Do not add any further functionality.

Save your code as **Q04aFINISHED** with the correct file extension for the programming language.

(7)

(b) Zak plans to implement a binary search algorithm to search a table of products.

(i) Explain **one** advantage of a binary search compared to a linear search.

(2)

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P 6 6 4 9 0 A 0 1 1 1 6

6. Question 4 · part b.iv · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part b.iv only.

(ii) **Figure 2** shows an algorithm for a binary search.

```
1     #     Initialise variables
2     SET productSearch TO " "
3     SET startPosition TO 1
4     SET midPosition TO 0
5     SET endPosition TO LENGTH(productList)
6     SET found TO False
7
8     #     Print prompts, take and check input from user
9     SEND "Enter the product code" TO DISPLAY
10    RECEIVE productSearch FROM (STRING) KEYBOARD
11    REPEAT
12          midPosition = (startPosition + endPosition) DIV 2
13          IF productList[midPosition] < productSearch THEN
14                startPosition = midPosition + 1
15          END IF
16          IF productList[midPosition] > productSearch THEN
17                endPosition = midPosition – 1
18          END IF
19    UNTIL productList[midPosition] = productSearch OR startPosition = endPosition
```

Figure 2

This binary search algorithm will be used to search the product list for the product code **str15**.

Complete the table to indicate the order in which the product codes will be examined by the algorithm.

Write the number 1 by the first product code to be examined, 2 by the second code to be examined, and so on.

(4)

Position in list	Product code	Order examined
1	ark11	
2	asp11	
3	bar13	
4	dri15	
5	mil19	
6	rib10	
7	str15	
8	tor16	

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P 6 6 4 9 0 A 0 1 2 1 6

6. Question 4 · part b.iv · 1 mark

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Further practice · Attempt part b.iv only.

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(iii) Zak has another list containing the names of five students who attend the after-school club.

Give the maximum number of names that would need to be examined by the binary search algorithm to determine whether a name appears in the list.

(1)

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(iv) Give the name of an algorithm that could be used to sort a list.

(1)

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(Total for Question 4 = 15 marks)

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Working and answer

7. Question 5 · part b.ii · 2 marks

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Further practice · Attempt part b.ii only.

5 Julia runs a computer gaming club.

(a) She wants a program to check passwords stored in a file.

The file **passwords.txt** contains the list of passwords.

The program must:

- check each password to ensure that:
  - the first character is an uppercase letter
  - if it is, that it also includes at least one digit (0–9)
- if a password does not meet these requirements:
  - display the password
  - increment the number of incorrect passwords
- display the total number of incorrect passwords after all the passwords have been checked.

Open **Q05a** in the code editor.

Write the program.

You must use the structure given in **Q05a** to write the program.

Do not add any further functionality.

Save your code as **Q05aFINISHED** with the correct file extension for the programming language.

(9)

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7. Question 5 · part b.ii · 2 marks

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Further practice · Attempt part b.ii only.

(b) **Figure 3** shows an array that stores player scores after a game.

3	2	10	8	1	9

Figure 3

(i) Julia uses a bubble sort algorithm to sort the scores.

Complete the table to show how the bubble sort algorithm will sort the scores.

You may not need to use all the rows.

(3)

3	2	10	8	1	9

(ii) Explain **one** reason why a bubble sort is very efficient in terms of memory usage.

(2)

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P 7 2 4 0 6 A 0 1 1 1 6

7. Question 5 · part b.ii · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part b.ii only.

(c) Describe the steps a linear search algorithm takes to find a search item.

(3)

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(Total for Question 5 = 17 marks)

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Working and answer

8. Question 5 · part c · 3 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.

5 Julia runs a computer gaming club.

(a) She wants a program to check passwords stored in a file.

The file **passwords.txt** contains the list of passwords.

The program must:

- check each password to ensure that:
  - the first character is an uppercase letter
  - if it is, that it also includes at least one digit (0–9)
- if a password does not meet these requirements:
  - display the password
  - increment the number of incorrect passwords
- display the total number of incorrect passwords after all the passwords have been checked.

Open **Q05a** in the code editor.

Write the program.

You must use the structure given in **Q05a** to write the program.

Do not add any further functionality.

Save your code as **Q05aFINISHED** with the correct file extension for the programming language.

(9)

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8. Question 5 · part c · 3 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.

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(b) **Figure 3** shows an array that stores player scores after a game.

3	2	10	8	1	9

Figure 3

(i) Julia uses a bubble sort algorithm to sort the scores.

Complete the table to show how the bubble sort algorithm will sort the scores.

You may not need to use all the rows.

(3)

3	2	10	8	1	9

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(ii) Explain **one** reason why a bubble sort is very efficient in terms of memory usage.

(2)

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P 7 2 4 0 6 A 0 1 1 1 6

8. Question 5 · part c · 3 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.

(c) Describe the steps a linear search algorithm takes to find a search item.

(3)

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(Total for Question 5 = 17 marks)

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Working and answer